A *wtt*-introimmune set in Π^0_1 and introimmunity for several reducibilities

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Abstract

We prove that there exists a weak truth-table introimmune set in the class Π^0_1, settling the question left open in previous work of whether the known Δ^0_2 existence result can be improved to Π^0_1. Since Σ^0_1 sets cannot be immune, this is best possible for weak truth-table introimmunity. We also study introimmunity for Jockusch's bounded-search reducibility $\leq_{bs}$ and Andersen's Dartmouth reducibility $\leq_D$, proving the existence of Δ^0_2 sets that are bs-introimmune and D-introimmune; hence there also exists a Δ^0_2 D^+-introimmune set. We next consider the classical reducibility $\leq_Q$, which is not contained in $\leq_T$ on all subsets of ω. We show that no infinite Π^0_1 set is Q-introimmune, while a Δ^0_2 Q-introimmune set does exist. Thus the existence of Δ^0_2 Q-introimmune sets is best possible within the arithmetical hierarchy. Finally, for enumeration reducibility $\leq_e$, we show that no infinite Π^1_1 set is e-introimmune, although e-introimmune sets do exist in the unrestricted sense. The proofs combine finite-injury priority arguments with dynamic spacing methods for $\leq_{wtt}$, $\leq_{bs}$, and $\leq_D$, a bit-by-bit finite-extension construction for $\leq_Q$, and an application of Soare's abstract existence theorem in the enumeration case.

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1 Introduction

The notion of introimmunity may be viewed as a strong opposite to introreducibility. Indeed, while a set fails to be introreducible as soon as one infinite subset does not compute it, introimmunity requires that no coinfinite subset computes the ambient set via the reducibility under consideration. Recent work on introreducibility and related notions may be found in [11].

The existence of sets without subsets of higher Turing degree was first established by Soare [16]. Concerning their definability, Jockusch proved that such sets cannot be arithmetical [13], and Simpson later strengthened this by showing that they cannot be hyperarithmetical [15]. Following [7], where this terminology was introduced in the polynomial-time setting, given a reducibility $\leq_r$, an infinite set $A \subseteq \omega$ is called *r-introimmune* if for every subset $B \subseteq A$ with $|A \setminus B| = \infty$ we have $A \not\leq_r B$. Whenever $\leq_r$ is transitive, r-introimmunity ensures that A has no subset of strictly higher r-degree. Moreover, if $\leq_r \Rightarrow \leq_s$, then every s-introimmune set is automatically r-introimmune.

A classical line of research asks how low in the arithmetical hierarchy such sets can occur for reducibilities below $\leq_T$. For the many-one reducibility $\leq_m$, it is known that m-introimmune sets exist in the class Π^0_1 [3, 4]. For truth-table type reducibilities, the existence of c-introimmune sets in

Δ^0_4 was proved in [7], where $\leq_c$ is the conjunctive reducibility, and Ambos-Spies[1] [1] later improved this to the existence of tt-introimmune sets in Δ^0_2. Further structural properties of degrees of sets having no subsets of higher m- and tt-degree were investigated in [6].

In [5], we turned to the weak truth-table reducibility $\leq_{wtt}$ (also called bounded Turing reducibility), proving the existence of wtt-introimmune sets in Δ^0_2. That paper left open whether the optimal arithmetical bound could be pushed down to Π^0_1.

A further motivation for the present work comes from the landscape of reducibilities below Turing reducibility. Earlier investigations naturally concentrated on transitive reducibilities, since only in that setting does one obtain a corresponding degree structure and can interpret introimmunity in degree-theoretic terms. However, the picture below $\leq_T$ is subtler than the standard chain

$$\leq_1 \Rightarrow \leq_m \Rightarrow \leq_{tt} \Rightarrow \leq_{wtt} \Rightarrow \leq_T$$

might suggest. The literature contains natural notions which are intermediate only on restricted classes, or which fail to be transitive and therefore do not induce degrees on all subsets of ω. This makes it necessary to distinguish carefully between the degree-theoretic reading of introimmunity and the more basic reducibility-theoretic condition $A \not\leq_r B$ for coinfinite subsets $B \subseteq A$. In particular, Andersen introduced the Dartmouth reducibilities $\leq_D$ and $\leq_{D^+}$, showing that $\leq_{D^+}$ is a transitive reducibility strictly below $\leq_T$ and incomparable with $\leq_{wtt}$ [2]; see also the corrigendum to [5] in [8]. This suggests enlarging the investigation from the usual wtt-chain to reducibilities which are natural in this context even when they do not carry a degree structure.

Our first main result is the existence of a wtt-introimmune set in Π^0_1. This gives the lowest possible arithmetical level for weak truth-table introimmunity. The main obstacle is that the Δ^0_2 construction of [5] relied on a K-computable dominating function to impose in advance sufficiently large gaps between successive elements of the set, thereby isolating the bounded uses of the relevant wtt-computations. In the present Π^0_1 setting the approximation is strictly subtractive, so such gaps cannot be precomputed. To overcome this, we develop a *dynamic spacing* technique: whenever the computable use of a convergent wtt-reduction threatens the next candidate marker, we permanently remove all intermediate elements. Because removals in a Π^0_1 construction are irreversible, this clearing process creates the required separation once and for all and makes the diagonalization possible within a finite-injury priority argument.

Further main results concern reducibilities lying outside the usual wtt-chain. In Section 3 we consider Jockusch's bounded-search reducibility $\leq_{bs}$ [12] and Andersen's Dartmouth reducibility $\leq_D$ [2]. Unlike wtt-reductions, neither bs-reductions nor D-reductions come equipped with a computable a priori use bound, so a purely subtractive Π^0_1 construction appears substantially more rigid for them. Nevertheless, by letting the halting problem dynamically compute the relevant spacing bounds, we prove the existence of Δ^0_2 sets that are bs-introimmune and D-introimmune. Since $\leq_{D^+} \Rightarrow \leq_D$, the latter result automatically yields a D^+-introimmune set as well. This leaves open whether the Δ^0_2 upper bound for $\leq_{bs}$ and $\leq_D$ can be lowered to Π^0_1.

Another part of the paper is devoted to the classical reducibility $\leq_Q$, defined by

$$A \leq_Q B \iff (\exists \text{ total computable } f)(\forall x)\,[x \in A \iff W_{f(x)} \subseteq B].$$

Although $\leq_Q$ does not belong to the framework of reducibilities strictly below $\leq_T$ on all subsets of ω, it is natural to include it in a broader study of arithmetical introimmunity. Indeed, $\leq_m \Rightarrow \leq_Q$, while $\leq_Q$ and $\leq_T$ are incomparable on 2^ω. On the class of computably enumerable sets one has $\leq_Q \Rightarrow \leq_T$,

[1] K. Ambos-Spies informed the author in private correspondence that the published argument for Theorem 3 of [1] requires correction, and very kindly communicated a corrected version.

but this is irrelevant here, since Q-introimmune sets must be immune. The behavior of introimmunity for $\leq_Q$ turns out to be sharply different from the reducibilities considered above: every Π^0_1 set A is Q-reducible to each subset $B \subseteq A$, so no infinite Π^0_1 set is Q-introimmune. On the other hand, in Section 4 we construct a Q-introimmune set in Δ^0_2. Thus the existence of Δ^0_2 Q-introimmune sets is best possible within the arithmetical hierarchy. Finally, in Section 5 we turn to enumeration reducibility $\leq_e$. Here the picture is sharply different from the arithmetical one described above. Using the characterization of Π^1_1 sets by uniform c.e. moduli from [11], we show that no infinite Π^1_1 set is e-introimmune; in particular, there are no arithmetical or hyperarithmetical e-introimmune sets. On the other hand, an application of Soare's abstract theorem for countable classes of Borel measurable partial maps yields the existence of e-introimmune sets in the unrestricted sense [16]. Thus enumeration reducibility admits introimmune sets, but only beyond the Π^1_1 level.

Taken together, the results of this paper yield the following picture. For the reducibilities below $\leq_T$ treated here, namely $\leq_{wtt}$, $\leq_{bs}$, $\leq_D$, and $\leq_{D^+}$, arithmetical introimmune sets do exist, and the optimal bound is obtained for $\leq_{wtt}$. The reducibility $\leq_Q$ also admits arithmetical introimmune sets, with the sharp bound Δ^0_2. By contrast, e-introimmune sets exist only abstractly: they do exist, but no infinite Π^1_1 set is e-introimmune. This considerably broadens the current existence picture for introimmune sets and highlights the descriptive diversity of the phenomenon across different reducibilities.

Conventions and notation. Our notation and basic background from computability theory are standard. For general references, see, e.g., Downey–Hirschfeldt [9], Odifreddi [14] and Soare [17].

Throughout the paper, sets are identified with their characteristic functions. We fix standard effective enumerations $\{W_e\}_{e\in\omega}$ of the computably enumerable sets, $\{\Phi_e\}_{e\in\omega}$ of oracle Turing functionals, and $\{\varphi_e\}_{e\in\omega}$ of partial computable functions. We write $K = \emptyset'$ for the halting set, use $\langle\cdot,\cdot\rangle$ for a fixed computable pairing function, and for an infinite set $X = \{x_0 < x_1 < \cdots\}$ let $p_X(n) = x_n$ denote its principal function.

2 A *wtt*-introimmune set in Π^0_1

Weak truth-table reducibility

In this section we settle the open question left in [5] by proving that weak truth-table introimmunity already occurs at the Π^0_1 level. The main difficulty is that, in a purely subtractive construction, the separation needed to control bounded uses cannot be fixed in advance and must instead be created dynamically during the construction.

Definition 2.1 (Weak truth-table reducibility)**.** Let $A, B \subseteq \omega$. We say that A is *weak truth-table reducible* to B, and write $A \leq_{wtt} B$, if there exist a Turing functional Φ and a total computable function g such that for every x,

$$\Phi^B(x)\downarrow = A(x) \text{ and every oracle query made in the computation } \Phi^B(x) \text{ is } \leq g(x).$$

Equivalently, $A \leq_{wtt} B$ iff there exist indices a, b such that φ_b is total and, for all x, $\Phi^B_a(x)\downarrow = A(x)$ with every oracle query in this computation $\leq \varphi_b(x)$. In this case we say that the pair (Φ_a, φ_b) is a *wtt*-reduction of A to B.

Remark 2.2. The reducibility $\leq_{wtt}$ is also known as *bounded Turing reducibility* $\leq_{bT}$.

We can now state and prove our main existence theorem.

Theorem 2.3. *There exists a wtt-introimmune set in* Π^0_1.

We construct a set A satisfying the following requirements for every $e = \langle a, b\rangle$:

$$N_e : (\Phi_a, \varphi_b) \text{ does not } \textit{wtt}\text{-reduce } A \text{ to any } X \subseteq A \text{ with } |A \setminus X| = \infty.$$

We do not need explicit positive requirements because the finite injury nature of our subtractive construction intrinsically guarantees that A is infinite.

Strategy and Construction The set A is constructed by stages $s \geq 0$. At each stage s we define a cofinite set A_s, and the final set will be

$$A = \bigcap_{s \geq 0} A_s.$$

Since $A_{s+1} \subseteq A_s$ for every $s \geq 0$ and the sequence $(A_s)_{s \geq 0}$ is uniformly computable, A is a Π^0_1 set. At any stage s, the elements of A_s are strictly ordered as

$$A_s = \{m^s_0 < m^s_1 < m^s_2 < \dots\}.$$

To satisfy the requirements, for each pair (n, e) with $e \leq n$ we maintain a finite set of strings $D^s_{n,e} \subseteq \{0,1\}^n$, representing the "defeated" oracle guesses.

Algorithm

- Stage 0. Set $A_0 = \omega$ (so $m^0_n = n$ for all $n \geq 0$). Set $D^0_{n,e} = \emptyset$ for all $e \leq n$.

- Stage $s+1$. Let A_s be the set constructed by the end of stage s. A pair (n, e) with $e \leq n \leq s$ and $e = \langle a, b\rangle$ *is eligible to act* at stage $s+1$ if $\varphi_{b,s}(m^s_k) \downarrow$ for all $k \leq n$. Let $u = \max_{k \leq n} \varphi_{b,s}(m^s_k)$ for this pair. The pair requires attention if one of the following two conditions holds:

  - **C1**: $A_s \cap (m^s_n, u] \neq \emptyset$ (which is equivalent to $m^s_{n+1} \leq u$).
  - **C2**: Condition **C1** does not hold, and there exists a string $\alpha \in \{0,1\}^n \setminus D^s_{n,e}$ such that, letting $X_{\alpha,s} \subseteq \omega$ be the *total* set defined by

$$X_{\alpha,s}(y) = \begin{cases} \alpha(k) & \text{if } y = m^s_k \text{ for some } k < n, \\ 0 & \text{otherwise,} \end{cases}$$

    we have $\Phi^{X_{\alpha,s}}_{a,s}(m^s_k) \downarrow = 1$ for all $k \leq n$, and every oracle query in these computations is $\leq u$ (hence the outcome depends only on $X_{\alpha,s} \upharpoonright (u+1)$).

  Find the lexicographically least pair (n, e) with $e \leq n \leq s$ that requires attention. If no such pair exists, set $A_{s+1} = A_s$, $D^{s+1}_{j,i} = D^s_{j,i}$ for all j, i, and go to the next stage. Otherwise, act according to the condition satisfied by (n, e):

  - **Action 1**: If (n, e) requires attention via **C1**, set $A_{s+1} = A_s \setminus (m^s_n, u]$. Set $D^{s+1}_{n,e} = D^s_{n,e}$. Let $y_{\min} = m^s_{n+1}$ be the minimum element extracted.
  - **Action 2**: If (n, e) requires attention via **C2** with some string α (pick the lexicographically least if there are many), set $A_{s+1} = A_s \setminus \{m^s_n\}$. Set $D^{s+1}_{n,e} = D^s_{n,e} \cup \{\alpha\}$. Let $y_{\min} = m^s_n$ be the minimum element extracted.

Reset Rule: For all pairs (j,i) with $j \geq 1$ and $i \leq j$ such that $m^s_{j-1} \geq y_{\min}$, set $D^{s+1}_{j,i} = \emptyset$. For all other pairs (j,i), if it is not the pair explicitly updated by Action 2, carry over $D^{s+1}_{j,i} = D^s_{j,i}$.

End of algorithm

Lemma 2.4. *Every marker m^s_n stabilizes, i.e., $\lim_{s\to\infty} m^s_n = m_n < \infty$. Consequently, A is infinite.*

Proof. We proceed by induction on n. For $n = 0$ the claim is immediate: the only relevant pair is $(0,0)$ and Action 2 can occur at most once since $\{0,1\}^0$ has size 1. Hence m^s_0 stabilizes. Now assume $n \geq 1$ and proceed by induction. Let s_0 be a stage such that the previous marker m^s_{n-1} has reached its final stable value m_{n-1}. This implies that for all $s \geq s_0$, no element less than or equal to m_{n-1} is ever extracted from A_s. Therefore, for any extraction at stage $s+1 > s_0$, the minimum extracted element $y_{\min}$ is strictly greater than m_{n-1}. By the Reset Rule, since $m_{n-1} < y_{\min}$, the set $D_{n,e}$ is never reset to $\emptyset$ after stage s_0 for any $e \leq n$.

We count how many times m^s_n can change (i.e., increase) after s_0. The value m^s_n increases if and only if an element $y \leq m^s_n$ is extracted. Since $y \geq y_{\min} > m_{n-1}$ (and A_s contains no elements strictly between m_{n-1} and m^s_n), the minimum element extracted by any action affecting m^s_n must be exactly m^s_n. Let us check which actions have $y_{\min} = m^s_n$:

- Action 1 for a pair (k,e) extracts elements with minimum m^s_{k+1}. This equals m^s_n if and only if $k = n-1$. Since $m_0, \ldots, m_{n-1}$ are stable, the combined use $u = \max_{j\leq n-1} \varphi_{b,s}(m_j)$ can transition from divergent to convergent at most once (because φ_b is a standard partial computable function, and once it halts on stable inputs, its value is fixed permanently). Thus Action 1 for $(n-1,e)$ clears the interval at most once for each $e \leq n-1$, after which Condition **C1** never holds again for $(n-1,e)$.

- Action 2 for a pair (k,e) extracts m^s_k. This equals m^s_n if and only if $k = n$. When Action 2 for (n,e) is executed, a string $\alpha \in \{0,1\}^n$ is permanently added to $D_{n,e}$. Since $D_{n,e}$ is never reset after s_0, and there are exactly 2^n strings of length n, this action can occur at most 2^n times for each $e \leq n$.

- Any other action (Action 1 for $k \geq n$, Action 2 for $k > n$) has a minimum extracted element strictly greater than m^s_n, leaving m^s_n unchanged.

- Action 2 for $k < n$ would extract $m_k \leq m_{n-1}$, which contradicts the stability of m_{n-1}, so it never happens after s_0.

- Similarly, Action 1 for $k < n-1$ would extract elements with minimum $m^s_{k+1} \leq m_{n-1}$, contradicting the stability of m_{n-1}, so it never happens after s_0.

Therefore, after s_0, the marker m^s_n is extracted at most $n + (n+1)2^n$ times. Hence $\lim_{s\to\infty} m^s_n$ exists and is finite. □

Lemma 2.5. *For every e, N_e is satisfied. Thus A is wtt-introimmune.*

Proof. Let $e = \langle a,b\rangle$. If φ_b is not a total computable function, then by definition the pair (Φ_a, φ_b) cannot be a valid *wtt*-reduction. In this case, the requirement N_e is vacuously satisfied. Therefore, from now on we may assume that φ_b is total.

For the sake of contradiction, assume that the requirement N_e is not satisfied. This implies that there exists an $X \subseteq A$ such that $|A \setminus X| = \infty$ and $\Phi^X_a = A$ with the queries bounded by the total computable function φ_b. Since X omits infinitely many elements of A, and $A = \{m_0 < m_1 < m_2 < \ldots\}$ is infinite by Lemma 2.4, there exists an integer $n \geq e$ such that $X(m_n) = 0$,

where $m_n = \lim_{s\to\infty} m_n^s$. Let s_0 be a sufficiently large stage such that for all $k \leq n+1$ and $s \geq s_0$, $m_k^s = m_k$. Since we have established that φ_b is total, the computation of $\varphi_b(m_k)$ must eventually converge for all $k \leq n$. Let $u = \max_{k\leq n} \varphi_b(m_k)$. Let $s_1 \geq s_0$ be a stage such that $\max_{k\leq n} \varphi_{b,s_1}(m_k) \downarrow = u$.

Since all markers up to m_{n+1} are stable from stage s_0, no pair lexicographically strictly less than (n, e) can require attention at any stage $s \geq s_1$, because any such pair would act by extracting an element $\leq m_{n+1}$, contradicting its stability. Hence, if **C1** held for (n, e) at some stage $s \geq s_1$, then (n, e) would be the highest priority pair requiring attention at that stage and would act via Action 1, extracting m_{n+1} and contradicting the stability of m_{n+1}. Therefore **C1** is false for all $s \geq s_1$, so the interval $(m_n, u]$ is completely empty in A and hence empty in $X \subseteq A$.

Let $\alpha \in \{0,1\}^n$ be the string defined by $\alpha(k) = X(m_k)$ for every $k < n$. Define the (total) set $X_\alpha \subseteq \omega$ by $X_\alpha(m_k) = \alpha(k)$ for $k < n$ and $X_\alpha(y) = 0$ for all other y. (At stage s we only use its restriction to $[0, u_s]$, where $u_s = \max_{k\leq n} \varphi_{b,s}(m_k^s)$.) We show that X_α matches the true oracle X up to u:

- For $y = m_k$ with $k < n$: $X_\alpha(y) = \alpha(k) = X(y)$.
- For $y = m_n$: $X_\alpha(y) = 0$ (by definition) and $X(y) = 0$ (by the choice of n).
- For all other $y \leq u$: $y \notin \{m_0, \ldots, m_n\}$, hence either $y < m_0$, or $m_{k-1} < y < m_k$ for some $1 \leq k \leq n$, or $y \in (m_n, u]$. In all cases $y \notin A$, hence $y \notin X$, so $X(y) = 0 = X_\alpha(y)$.

Therefore, X exactly coincides with X_α up to the use bound u. Since we are assuming $\Phi_a^X = A$ and $m_k \in A$ for all $k \leq n$, the true computation $\Phi_a^X(m_k)$ must output 1. Since the true computation $\Phi_a^X(m_k)$ only queries elements $\leq \varphi_b(m_k) \leq u$ and X agrees with X_α up to u, the simulated computation $\Phi_a^{X_\alpha}(m_k)$ is identical. Thus, it also outputs 1 and only queries elements $\leq u$, fulfilling the condition **C2**. Hence, the computation predicate in **C2** is satisfied via α for (n, e) for all sufficiently large stages $s \geq s_1$.

Fix a stage $s_2 \geq s_1$ large enough that the computation predicate in **C2** is witnessed at stage s_2 via α. Then, if $\alpha \notin D_{n,e}^{s_2}$, the pair (n, e) would act via Action 2 at stage $s_2 + 1$, contradicting the stability of m_n. Hence, $\alpha \in D_{n,e}^{s_2}$.

Let $t < s_2$ be such that $\alpha \in D_{n,e}^{t+1} \setminus D_{n,e}^t$ and t is maximal with this property. Then stage $t+1$ executed Action 2 for (n, e) via α, removing

$$x := m_n^t$$

from A_{t+1}.

Since $\alpha \in D_{n,e}^{s_2}$ and t was chosen maximal, $D_{n,e}$ is not reset at any stage between $t+1$ and s_2. Hence no stage between $t+1$ and s_2 extracts an element $\leq m_{n-1}^t$, so the markers $m_0^t, \ldots, m_{n-1}^t$ are already their final values $m_0, \ldots, m_{n-1}$. Therefore, α correctly codes X on these elements.

Because x is removed at stage $t+1$, we have $x \notin A$, so $A(x) = 0$, and also $X(x) = 0$ since $X \subseteq A$. Moreover, since Action 2 applies at stage $t+1$, condition **C1** evaluated on A_t must be false. Let

$$u_t := \max_{k\leq n} \varphi_{b,t}(m_k^t).$$

Then $A_t \cap (x, u_t] = \emptyset$. Furthermore, by the definition of the markers, A_t contains no elements strictly between m_{k-1}^t and m_k^t for any $k \leq n$. Since $X \subseteq A \subseteq A_t$, it follows that X has no elements in $(x, u_t]$ nor strictly between any markers up to x. Thus, X exactly agrees with X_α up to u_t. Finally, since **C2** holds at stage $t+1$ via α, we have $\Phi_{a,t}^{X_\alpha}(x) \downarrow = 1$. (Notice that since $m_k^t = m_k$ for $k < n$, the evaluated set $X_{\alpha,t}$ exactly coincides with the X_α defined above, so we have $\Phi_{a,t}^{X_\alpha}(x) \downarrow = 1$). Since the

use of $\Phi_a^X(x)$ is $\leq \varphi_b(x) \leq u_t$ and X agrees with X_α below u_t, it follows that $\Phi_a^X(x) = \Phi_a^{X_\alpha}(x) = 1$, contradicting $A(x) = 0$. □

Proof of Theorem 2.3. By Lemma 2.4, the constructed set A is infinite. By Lemma 2.5, every requirement N_e is satisfied. Hence A is *wtt*-introimmune. □

3 Introimmunity for bounded-search and Dartmouth reducibilities in Δ_2^0

We now turn to two further reducibilities strictly below $\leq_T$, namely Jockusch's bounded-search reducibility and Andersen's Dartmouth reducibility. Since neither of them comes with a computable a priori bound on the relevant oracle use, the Π_1^0 strategy from the weak truth-table case must be replaced by a Δ_2^0 construction in which the necessary spacing is computed dynamically from the halting problem.

Definition 3.1 (Bounded-search and Dartmouth reducibilities)**.** Let $A, B \subseteq \omega$. For each index e, and input n, if $\Phi_e^B(n)\downarrow$, let $Q_e^B(n)$ denote the finite set of all numbers u such that the membership or non-membership of u in B is used in the computation of $\Phi_e^B(n)$.

(i) $A \leq_{bs} B$ if there is a Turing functional Φ_a and a total computable function φ_b such that $\Phi_a^B = A$ and
$$|Q_a^B(x)| \leq \varphi_b(x) \qquad \text{for all } x.$$

(ii) $A \leq_D B$ if there is a Turing functional Φ_a such that $\Phi_a^B = A$ and Φ_a^W is a total function for every computably enumerable set W.

(iii) $A \leq_{D^+} B$ if $A \leq_D B$ and the reduction Φ_a is a positive procedure. Of course $\leq_{D^+} \Rightarrow \leq_D$, and as shown by Andersen [2], $\leq_D \not\Rightarrow \leq_{D^+}$. Intuitively, a positive procedure uses only positive information from the oracle to certify output 1, and only negative information to certify output 0.

Remark 3.2. We may, and do, fix once and for all an effective enumeration of oracle Turing functionals that never repeat an oracle query. Indeed, from any oracle functional one can uniformly obtain an equivalent one that stores previously obtained oracle answers and reuses them whenever the same query is requested again. Hence, for functionals in this normalized enumeration, the total number of oracle queries made in the computation of $\Phi_a^B(x)$ is exactly $|Q_a^B(x)|$. Accordingly, in the proof below, whenever we say that a computation makes at most $\varphi_b(x)$ oracle queries, we are simply using the bound $|Q_a^B(x)| \leq \varphi_b(x)$ for the normalized functional.

Unlike *wtt*-reductions, the uses of *bs*-reductions and *D*-reductions are not bounded by an a priori computable function. This makes a purely subtractive Π_1^0 construction highly problematic. However, by using the halting set $K \equiv_T \emptyset'$ as an oracle, we can dynamically construct the topological spacing required for the diagonalization using a finite extension argument, obtaining $\{bs, D\}$-introimmune sets in Δ_2^0.

Theorem 3.3. *There exists a set $A \in \Delta_2^0$ which is bs-introimmune.*

Proof. We construct $A \in \Delta^0_2$ by finite extensions, satisfying the following requirements for all $a, b, e \in \omega$:

$$P_{2e} : |A| \geq e,$$
$$N_{2\langle a,b\rangle+1} : (\Phi_a, \varphi_b) \text{ does not } bs\text{-reduce } A \text{ to any } X \subseteq A \text{ with } |A \setminus X| = \infty.$$

Strategy and Dynamic Spacing. The set A will be constructed by stages $s = 0, 1, \ldots$. At stage s, we define a finite string $\alpha_s \in \{0,1\}^s$, letting $A_s = X_{\alpha_s} = \{h(n) : n < s \wedge \alpha_s(n) = 1\}$, where h is a strictly increasing function. The final set is $A = \bigcup_s A_s$.

Unlike the $\leq_{wtt}$ case, the use of bs-reductions cannot be bounded a priori by a single total computable function. To overcome this, we define $h(s)$ *dynamically* using the halting set $K \equiv_T \emptyset'$.

Given a negative requirement N_c where $c = 2\langle a, b\rangle + 1$ and an input x, we use K to compute a K-computable upper bound $U(c, x)$ on the use of any computation of $\Phi_a(x)$ making at most $\varphi_b(x)$ oracle queries. Specifically, we ask K whether $\varphi_b(x) \downarrow$. If not, set $U(c, x) = 0$. If it converges, let $k = \varphi_b(x)$. For each $\sigma \in \{0,1\}^{\leq k}$, we simulate $\Phi_a(x)$ by feeding the successive oracle answers prescribed by σ. Using K, we decide whether this simulation halts after asking at most $|\sigma|$ queries. Let $U(c, x)$ be the maximum query asked in any such halting simulation (or 0 if there is none). Moreover, if c is even we conventionally set $U(c, x) = 0$ for every x.

We define the strictly increasing sequence $h(n)$ recursively:

$$h(0) = 0, \quad h(s+1) = \max\Big(\{h(s)\} \cup \{U(c, h(m)) : c \leq s,\ m \leq s\}\Big) + 1.$$

Because U is K-computable, the sequence h is K-computable. This definition enforces the following crucial structural bound:

Claim 3.4. *Let $c = 2\langle a, b\rangle + 1$ be odd. If $s \geq \max(c, m)$ and for some oracle F the computation $\Phi_a^F(h(m))$ halts making at most $\varphi_b(h(m))$ queries, then every query asked in this computation is $< h(s+1)$.*

Proof of Claim 3.4. Let $k = \varphi_b(h(m))$. The actual successive oracle answers seen in the computation form some $\sigma \in \{0,1\}^{\leq k}$. By definition of $U(c, h(m))$, every query asked in the computation is at most $U(c, h(m))$. Since $c \leq s$ and $m \leq s$, the recursive definition of h gives $U(c, h(m)) < h(s+1)$. Hence every query asked in the computation is $< h(s+1)$. □

Construction. Given any string α, let $X_\alpha = \{h(n) : n < |\alpha| \wedge \alpha(n) = 1\}$. Fix a stage $s+1$, having constructed α_s of length s. Requirement P_{2e} requires attention if $|X_{\alpha_s}| < e$. Requirement N_c (with $c \leq s$) requires attention via a string $\alpha \in \{0,1\}^s$ if:

- **C1** $\alpha(m) \leq \alpha_s(m)$ for all $m < s$ (i.e., $X_\alpha \subseteq X_{\alpha_s}$);
- **C2** for every $m < s$, $\Phi_a^{X_\alpha}(h(m)) \downarrow = \alpha_s(m)$ and the computation halts making $\leq \varphi_b(h(m))$ queries;
- **C3** $\Phi_a^{X_{\alpha 0}}(h(s)) \downarrow = 1$ and the computation halts making $\leq \varphi_b(h(s))$ queries.

A requirement is *active* if it is the highest priority requirement requiring attention.

- **Stage 0:** $\alpha_0 = \emptyset$.
- **Stage $s+1$:** Use K to compute $h(s+1)$ and let R_n be the least requirement with $n \leq 2(s+1)$ that requires attention at stage $s+1$. Such an n exists, since $P_{2(s+1)}$ requires attention: indeed, $|X_{\alpha_s}| \leq s < s+1$. If n is even, set $\alpha_{s+1} = \alpha_s 1$; if n is odd, set $\alpha_{s+1} = \alpha_s 0$.

We set $A = \bigcup_s X_{\alpha_s}$. Since the search for active requirements and the generation of $h(s)$ only require finitely many queries to K, $A \leq_T K$, so $A \in \Delta^0_2$.

Verification. We prove by induction on the priority that every requirement acts only finitely often and is met.

First let $R_n = P_{2e}$ be even, and assume inductively that every requirement R_d with $d < n$ acts only finitely often. Choose a stage $s_0 > n$ such that no requirement R_d with $d < n$ acts after stage s_0. If $|X_{\alpha_{s_0}}| \geq e$, then P_{2e} is already satisfied. Otherwise, at every later stage $t+1$ with $|X_{\alpha_t}| < e$, the requirement P_{2e} requires attention. Since no smaller-index requirement acts after s_0, it is active whenever it requires attention, so each such action appends a 1 and increases $|X_{\alpha_t}|$ by one. Therefore P_{2e} acts only finitely often and is eventually satisfied.

Now fix an odd requirement N_c, say $c = 2\langle a, b\rangle + 1$, and assume inductively that every requirement R_d with $d < c$ acts only finitely often. Choose a stage $s_0 > c$ such that no requirement R_d with $d < c$ acts after stage s_0.

For each $t \in \omega$, let V_t^c be the set of all strings $\beta \in \{0,1\}^t$ such that:

(V1) $\beta(m) \leq \alpha_t(m)$ for all $m < t$;

(V2) for every $m < t$, $\Phi_a^{X_\beta}(h(m)) \downarrow = \alpha_t(m)$ and the computation halts making $\leq \varphi_b(h(m))$ queries.

Thus N_c requires attention at stage $t+1$ via β if and only if $\beta \in V_t^c$ and

$$\Phi_a^{X_{\beta 0}}(h(t)) \downarrow = 1$$

with at most $\varphi_b(h(t))$ queries.

Claim 3.5. *If $s_0 \leq s < t$ and $\beta \in V_t^c$, then $\beta{\restriction}s \in V_s^c$.*

Proof. Condition (V1) is immediate because α_t extends α_s.

Fix $m < s$. Since $\beta \in V_t^c$, the computation $\Phi_a^{X_\beta}(h(m))$ halts making at most $\varphi_b(h(m))$ queries. Because $s \geq s_0 > c$ and $m < s$, we have $\max(c, m) \leq s - 1$. Claim 3.4, applied with $s-1$ in place of s, shows that every query asked in this computation is $< h(s)$.

Now $X_{\beta{\restriction}s}$ and X_β agree below $h(s)$, so

$$\Phi_a^{X_{\beta{\restriction}s}}(h(m)) = \Phi_a^{X_\beta}(h(m)) = \alpha_t(m) = \alpha_s(m).$$

Thus (V2) holds as well, and $\beta{\restriction}s \in V_s^c$. □

Claim 3.6. *Let $s \geq s_0$, let $\alpha \in V_s^c$, and suppose that $\beta \in V_t^c$ extends α for some $t > s$. Then*

$$\beta(s) = \alpha_{s+1}(s).$$

Proof. If $\alpha_{s+1}(s) = 0$, then $\alpha_t(s) = 0$ for every $t > s$, and since $\beta \in V_t^c$, condition (V1) yields

$$\beta(s) \leq \alpha_t(s) = 0.$$

Hence $\beta(s) = 0$.

Suppose now that $\alpha_{s+1}(s) = 1$, and assume for contradiction that $\beta(s) = 0$. Since $\beta \in V_t^c$, condition (V2) at $m = s$ gives

$$\Phi_a^{X_\beta}(h(s)) \downarrow = \alpha_t(s) = 1$$

with at most $\varphi_b(h(s))$ queries. Because $s \geq s_0 > c$, Claim 3.4 shows that every query asked in this computation is $< h(s+1)$.

Since β extends α and $\beta(s) = 0$, the sets X_β and $X_{\alpha 0}$ agree below $h(s+1)$. Therefore

$$\Phi_a^{X_{\alpha 0}}(h(s)) \downarrow = 1$$

with at most $\varphi_b(h(s))$ queries. As $\alpha \in V_s^c$, this means that α satisfies **C1**–**C3** at stage $s+1$, so N_c requires attention at stage $s+1$.

If some requirement R_d with $d < c$ required attention at stage $s+1$, then by priority it would act at stage $s+1$, contradicting the choice of s_0. Hence no such R_d requires attention, and therefore N_c is active at stage $s+1$. This forces $\alpha_{s+1}(s) = 0$, contradicting our assumption that $\alpha_{s+1}(s) = 1$.

Therefore $\beta(s) = 1$, as required. □

Claim 3.7. *If N_c acts at stage $s+1$ via some $\alpha \in V_s^c$, then no $\beta \in V_t^c$ with $t > s$ extends α.*

Proof. Since N_c acts at stage $s+1$, we have $\alpha_{s+1}(s) = 0$, and therefore $\alpha_t(s) = 0$ for every $t > s$.

Suppose that $\beta \in V_t^c$ extends α for some $t > s$. By (V1),

$$\beta(s) \leq \alpha_t(s) = 0,$$

so $\beta(s) = 0$.

Because N_c acted via α, condition **C3** held at stage $s+1$, i.e.

$$\Phi_a^{X_{\alpha 0}}(h(s)) \downarrow = 1$$

with at most $\varphi_b(h(s))$ queries. Again Claim 3.4 shows that every query asked in this computation is $< h(s+1)$.

Since β extends α and $\beta(s) = 0$, the sets X_β and $X_{\alpha 0}$ agree below $h(s+1)$. Hence

$$\Phi_a^{X_\beta}(h(s)) = \Phi_a^{X_{\alpha 0}}(h(s)) = 1.$$

But because $\beta \in V_t^c$, condition (V2) at $m = s$ requires

$$\Phi_a^{X_\beta}(h(s)) = \alpha_t(s) = 0,$$

a contradiction. Therefore no such β exists. □

Claims 3.5 and 3.6 imply that for every $t \geq s_0$, the restriction map

$$\beta \mapsto \beta{\restriction}t$$

sends V_{t+1}^c injectively into V_t^c. Moreover, if N_c acts at stage $t+1$, then Claim 3.7 shows that some element of V_t^c has no extension in V_{t+1}^c. Hence

$$|V_{t+1}^c| < |V_t^c|$$

whenever N_c acts at stage $t+1$, while always $|V_{t+1}^c| \leq |V_t^c|$ for $t \geq s_0$. Since $|V_{s_0}^c| \leq 2^{s_0}$, it follows that N_c acts only finitely often.

It remains to show that N_c is met. Suppose for contradiction that there exists $B \subseteq A$ with $|A \setminus B| = \infty$ such that $\Phi_a^B = A$ and, for every x, the computation $\Phi_a^B(x)$ makes at most $\varphi_b(x)$ queries.

Because $A \setminus B$ is infinite and $A \subseteq \{h(n) : n \in \omega\}$, we may choose $s \geq s_0$ such that

$$A(h(s)) = 1 \qquad \text{and} \qquad B(h(s)) = 0.$$

Define $\alpha \in \{0,1\}^s$ by

$$\alpha(m) = B(h(m)) \qquad (m < s).$$

We claim that $\alpha \in V_s^c$. First, since $B \subseteq A$ and $\alpha_s(m) = A(h(m))$ for all $m < s$, we have

$$\alpha(m) \leq \alpha_s(m) \qquad (m < s),$$

so (V1) holds.

Now fix $m < s$. Since $\Phi_a^B = A$, we have

$$\Phi_a^B(h(m)) = A(h(m)) = \alpha_s(m),$$

and this computation makes at most $\varphi_b(h(m))$ queries. Because $s \geq s_0 > c$ and $m < s$, Claim 3.4, applied with $s-1$ in place of s, shows that every query asked in this computation is $< h(s)$.

Moreover, because $B \subseteq A \subseteq \{h(n) : n \in \omega\}$ and $\alpha(m) = B(h(m))$ for $m < s$, we have

$$B \cap [0, h(s)) = X_\alpha \cap [0, h(s)).$$

Therefore

$$\Phi_a^{X_\alpha}(h(m)) = \Phi_a^B(h(m)) = \alpha_s(m),$$

with at most $\varphi_b(h(m))$ queries. Thus (V2) holds, and so $\alpha \in V_s^c$.

Finally,

$$\Phi_a^B(h(s)) = A(h(s)) = 1,$$

again with at most $\varphi_b(h(s))$ queries. Claim 3.4 shows that every query asked in this computation is $< h(s+1)$. Since $B(h(s)) = 0$ and $B \subseteq A \subseteq \{h(n) : n \in \omega\}$, we have

$$B \cap [0, h(s+1)) = X_{\alpha 0} \cap [0, h(s+1)).$$

Hence

$$\Phi_a^{X_{\alpha 0}}(h(s)) \downarrow = 1$$

with at most $\varphi_b(h(s))$ queries. Therefore α witnesses that N_c requires attention at stage $s+1$.

As before, no requirement R_d with $d < c$ can require attention at stage $s+1$, for otherwise it would act after stage s_0. Thus N_c is active at stage $s+1$ and acts, forcing $\alpha_{s+1}(s) = 0$. Consequently every later $\alpha_t(s)$ is also 0, and therefore $A(h(s)) = 0$, contradicting the choice of s.

This contradiction shows that no such set B exists. Hence N_c is met.

By induction on the priority, every requirement is met. Therefore A is bs-introimmune. □

Theorem 3.8. *There exists a set $A \in \Delta_2^0$ which is D-introimmune.*

(Since $\leq_{D^+} \Rightarrow \leq_D$, the set A is automatically D^+-introimmune as well).

Proof. We construct $A \in \Delta_2^0$ by finite extensions, satisfying the following requirements for all $a, e \in \omega$:

$$P_{2e} : |A| \geq e,$$
$$N_{2a+1} : \Phi_a \text{ does not } D\text{-reduce } A \text{ to any } X \subseteq A \text{ with } |A \setminus X| = \infty.$$

Strategy and Dynamic Spacing. The set A will be constructed by stages $s = 0, 1, \ldots$. At stage s, we define a finite string $\alpha_s \in \{0,1\}^s$, letting $A_s = X_{\alpha_s} = \{h(n) : n < s \wedge \alpha_s(n) = 1\}$, where h is a strictly increasing function. The final set is $A = \bigcup_s A_s$.

Unlike the $\leq_{wtt}$ case, the use of D-reductions cannot be bounded a priori by a single total computable function. To overcome this, we define $h(s)$ *dynamically* using the halting set $K \equiv_T \emptyset'$. Given a negative requirement N_c where $c = 2a + 1$, an input x, and a current bound y, we use K to compute the maximum possible use $U(c, x, y)$ of the reduction over all valid partial oracle configurations bounded by y. A valid D-reduction must be total on all c.e. sets. Since every finite set is c.e., $\Phi_a^F(x)$ must converge for every finite set F. There are exactly 2^{y+1} possible subsets $F \subseteq \{0, \ldots, y\}$. Using K, we check if $\Phi_a^F(x) \downarrow$ for *all* such subsets (assuming the oracle answers 0 for any query $> y$). If it diverges for at least one, Φ_a is not a valid D-reduction, we set $U(c, x, y) = 0$, and requirement N_c is permanently disabled. If it converges for all of them, let $U(c, x, y)$ be the maximum use among these 2^{y+1} halting computations (we always set $U(c, x, y) = 0$ if c is even). We define the strictly increasing sequence $h(n)$ recursively:

$$h(0) = 0, \quad h(s+1) = \max\Big(\{h(s)\} \cup \{U(c, h(m), h(s)) : c \leq s, m \leq s\}\Big) + 1.$$

Because U is K-computable, the sequence h is K-computable. This definition enforces the following crucial structural bound:

Claim 3.9. *If N_c defines a valid D-reduction, then for every $m \in \omega$ and every stage $s \geq \max(c, m)$, the use of the computation on input $h(m)$ with any finite oracle $F \subseteq \{h(0), \ldots, h(s)\}$ is strictly less than $h(s+1)$.*

Proof of Claim 3.9. Assume that N_c defines a valid D-reduction, and fix m and $s \geq \max(c, m)$. Let

$$F \subseteq \{h(0), \ldots, h(s)\}$$

be finite. Since h is strictly increasing, we have

$$F \subseteq \{0, \ldots, h(s)\}.$$

Because Φ_a is a valid D-reduction, it is total on every c.e. oracle. In particular, for every finite set

$$E \subseteq \{0, \ldots, h(s)\},$$

the computation $\Phi_a^E(h(m))$ converges. Hence, in the definition of $U(c, h(m), h(s))$, the disabling case does not occur, and

$$U(c, h(m), h(s))$$

is exactly the maximum use among the computations $\Phi_a^E(h(m))$ for all subsets $E \subseteq \{0, \ldots, h(s)\}$.

Now our given set F is one of these subsets. Therefore the use of $\Phi_a^F(h(m))$ is at most $U(c, h(m), h(s))$.

Finally, since $s \geq \max(c, m)$, we have $c \leq s$ and $m \leq s$, so the quantity $U(c, h(m), h(s))$ appears among the terms used to define $h(s+1)$. Thus

$$U(c, h(m), h(s)) < h(s+1).$$

It follows that the use of $\Phi_a^F(h(m))$ is strictly less than $h(s+1)$, as required. □

Construction. Given any string α, let $X_\alpha = \{h(n) : n < |\alpha| \wedge \alpha(n) = 1\}$. Fix a stage $s+1$, having constructed α_s of length s. Requirement P_{2e} requires attention if $|X_{\alpha_s}| < e$. Requirement N_c (with $c < s$, and not permanently disabled) requires attention via a string $\alpha \in \{0,1\}^s$ if:

C1 for every $m < s$, $\alpha(m) \leq \alpha_s(m)$;

C2 for every $m < s$, $\Phi_a^{X_\alpha}(h(m)) \downarrow = \alpha_s(m)$;

C3 $\Phi_a^{X_{\alpha 0}}(h(s)) \downarrow = 1$.

A requirement is *active* if it is the highest priority requirement requiring attention.

- **Stage 0:** $\alpha_0 = \emptyset$.
- **Stage $s+1$:** Use K to compute $h(s+1)$. Let R_n be the active requirement, if there is one. If no requirement requires attention, set $\alpha_{s+1} = \alpha_s 1$. If $R_n = P_{2e}$, set $\alpha_{s+1} = \alpha_s 1$. If $R_n = N_c$, then N_c requires attention via some string $\alpha \in \{0,1\}^s$, and we act by setting $\alpha_{s+1} = \alpha_s 0$.

We set $A = \bigcup_s X_{\alpha_s}$. Since at each stage the search for active requirements and the computation of $h(s+1)$ require only finitely many queries to K, the construction is K-computable. Hence $A \leq_T K$, and therefore $A \in \Delta_2^0$.

Verification. We verify by induction on the priority ordering that each requirement acts only finitely often and is satisfied.

Fix e. Let s_0 be a stage after which no requirement of higher priority than P_{2e} acts. If $|A_{s_0}| \geq e$, then P_{2e} is already met. Otherwise, for every stage $s \geq s_0$ with $|A_s| < e$, the requirement P_{2e} is the active requirement, so we set $\alpha_{s+1} = \alpha_s 1$. Each such action increases $|A_s|$ by one, and later stages never remove elements. Hence after finitely many stages we obtain $|A_s| \geq e$, and P_{2e} is permanently satisfied.

Now fix $c = 2a+1$. Let $s_0 > c$ be a stage after which no requirement of higher priority than N_c acts. If during the computation of some $U(c, h(m), h(s))$ we find a finite oracle F such that $\Phi_a^F(h(m)) \uparrow$, then N_c is permanently disabled, and there is nothing to prove. So assume this never happens. Then for every $t \geq s_0$, every $m < t$, and every finite oracle $F \subseteq \{h(0), \ldots, h(t-1)\}$, the computation $\Phi_a^F(h(m))$ has use $< h(t)$, by the definition of $U(c, h(m), h(t-1))$ and of $h(t)$.

For $t \geq s_0$, call a string $\gamma \in \{0,1\}^t$ *viable at stage t* if it satisfies **C1** and **C2** with respect to α_t.

If $\beta \in \{0,1\}^{t+1}$ is viable at stage $t+1$ and $\gamma = \beta \upharpoonright t$, then γ is viable at stage t. Indeed, **C1** is immediate. For **C2**, fix $m < t$. By the use bound above, the computation $\Phi_a^{X_\gamma}(h(m))$ has use $< h(t)$. Since X_β and X_γ coincide below $h(t)$, we get

$$\Phi_a^{X_\gamma}(h(m)) = \Phi_a^{X_\beta}(h(m)) = \alpha_{t+1}(m) = \alpha_t(m).$$

Thus γ is viable.

Now let $\gamma \in \{0,1\}^t$ be viable at stage t. We show that γ has at most one viable extension to length $t+1$.

If $\alpha_{t+1}(t) = 0$, then any viable extension β of γ must satisfy $\beta(t) \leq \alpha_{t+1}(t) = 0$ by **C1**, so necessarily $\beta = \gamma 0$.

If $\alpha_{t+1}(t) = 1$, then $\gamma 0$ cannot be viable at stage $t+1$. For if it were, since $X_{\gamma 0} = X_\gamma$, the string γ would satisfy **C1** and **C2** at stage t, and moreover $\Phi_a^{X_{\gamma 0}}(h(t)) = 1$. Hence N_c would require attention via γ at stage $t+1$. But after stage s_0 no higher priority requirement acts, so N_c would

then be active and we would have $\alpha_{t+1}(t) = 0$, a contradiction. Therefore in this case the only possible viable extension is $\gamma 1$.

Let v_t be the number of viable strings of length t. The previous paragraphs show that every viable string of length $t+1$ extends a viable string of length t, and that each viable string of length t has at most one viable extension. Therefore

$$v_{t+1} \leq v_t \qquad \text{for every } t \geq s_0.$$

Moreover, if N_c acts at stage $t+1 > s_0$ via a viable string $\gamma \in \{0,1\}^t$, then $\alpha_{t+1}(t) = 0$. Hence $\gamma 1$ fails **C1**, while $\gamma 0$ fails **C2** at input $h(t)$, because **C3** gives

$$\Phi_a^{X_{\gamma 0}}(h(t)) = 1 \neq \alpha_{t+1}(t) = 0.$$

So γ has no viable extension at stage $t+1$, and thus $v_{t+1} < v_t$. Since $v_{s_0} \leq 2^{s_0}$, the requirement N_c can act at most 2^{s_0} times after stage s_0. In particular, N_c acts only finitely often.

We now show that N_c is met. Suppose toward a contradiction that Φ_a D-reduces A to some set $B \subseteq A$ with $|A \setminus B| = \infty$. Choose $s \geq s_0$ such that

$$A(h(s)) = 1 \qquad \text{and} \qquad B(h(s)) = 0.$$

Define $\alpha \in \{0,1\}^s$ by

$$\alpha(m) = B(h(m)) \qquad (m < s).$$

We claim that N_c requires attention via α at stage $s+1$.

Condition **C1** holds because $B \subseteq A$, so for every $m < s$,

$$\alpha(m) = B(h(m)) \leq A(h(m)) = \alpha_s(m).$$

To verify **C2**, fix $m < s$. Since $X_\alpha \subseteq \{h(0), \ldots, h(s-1)\}$, Claim 3.9 applied at stage $s-1$ gives that the computation $\Phi_a^{X_\alpha}(h(m))$ has use $< h(s)$. Also, because $B \subseteq A \subseteq \text{range}(h)$, the sets X_α and B coincide below $h(s)$. Therefore

$$\Phi_a^{X_\alpha}(h(m)) = \Phi_a^{B}(h(m)) = A(h(m)) = \alpha_s(m).$$

Finally, to verify **C3**, note that $B(h(s)) = 0$. Hence $X_{\alpha 0}$ and B coincide below $h(s+1)$: below $h(s)$ this follows from the definition of α, at $h(s)$ both answers are 0, and any element of B above $h(s)$ must be equal to $h(j)$ for some $j \geq s+1$, hence is at least $h(s+1)$. By Claim 3.9 applied at stage s, the computation $\Phi_a^{X_{\alpha 0}}(h(s))$ has use $< h(s+1)$. Thus

$$\Phi_a^{X_{\alpha 0}}(h(s)) = \Phi_a^{B}(h(s)) = A(h(s)) = 1.$$

So α satisfies **C1**–**C3**, and therefore N_c requires attention at stage $s+1$.

Since no higher priority requirement acts after s_0, it follows that N_c acts at stage $s+1$. But then $\alpha_{s+1}(s) = 0$, i.e. $A(h(s)) = 0$, contradicting the choice of s. This contradiction shows that no such set B exists. Hence N_c is met.

By induction, every requirement is met. Therefore A is D-introimmune.

□

4 Arithmetical Q-introimmune sets

This section is devoted to the classical reducibility $\leq_Q$, which falls outside the framework of reducibilities strictly below $\leq_T$ on all subsets of ω. Here the situation can be determined exactly at the level of arithmetical existence: infinite Π^0_1 sets are never Q-introimmune, whereas Δ^0_2 Q-introimmune sets do exist.

Definition 4.1 (Q-reducibility)**.** Let $A, B \subseteq \omega$. We say that $A \leq_Q B$ if there exists a total computable function f such that

$$x \in A \iff W_{f(x)} \subseteq B \qquad \text{for all } x \in \omega.$$

Theorem 4.2. *Let $A \subseteq \omega$ be a Π^0_1 set. Then $A \leq_Q B$ for every $B \subseteq A$. In particular, no infinite Π^0_1 set is Q-introimmune.*

Proof. Since A is Π^0_1, its complement $\overline{A}$ is c.e. Fix an index e such that

$$W_e = \overline{A}.$$

By the *s-m-n* theorem, there exists a total computable function f such that, for each $x \in \omega$, the c.e. set $W_{f(x)}$ is generated by the program which waits for x to appear in the enumeration of W_e and, if this happens, enumerates x. Hence

$$W_{f(x)} = \begin{cases} \varnothing & \text{if } x \in A, \\ \{x\} & \text{if } x \notin A. \end{cases}$$

Now let $B \subseteq A$. If $x \in A$, then $W_{f(x)} = \varnothing \subseteq B$. If $x \notin A$, then $W_{f(x)} = \{x\}$, and since $B \subseteq A$, we have $x \notin B$, so $W_{f(x)} \nsubseteq B$. Therefore,

$$x \in A \iff W_{f(x)} \subseteq B.$$

Since f is total computable, this proves that $A \leq_Q B$.

For the final claim, assume that A is infinite and choose $B \subseteq A$ so that both B and $A \setminus B$ are infinite. Then $A \leq_Q B$ by the first part, and hence A is not Q-introimmune. □

Theorem 4.3. *There exists a set $A \in \Delta^0_2$ which is Q-introimmune.*

Proof. We construct $A \in \Delta^0_2$ by finite extensions, using the halting set $K \equiv_T \emptyset'$ as an oracle. Let $\{V_{e,x}\}_{e,x \in \omega}$ be a standard effective enumeration of all uniformly computably enumerable families of sets. It suffices to satisfy the following requirements for all $e \in \omega$:

$$P_e : |A| \geq e,$$
$$R_e : \text{The sequence } V_{e,x} \text{ does not } Q\text{-reduce } A \text{ to any } B \subseteq A \text{ with } |A \setminus B| = \infty.$$

To prevent lower-priority requirements from jumping over elements and bypassing conditions, the construction proceeds *bit-by-bit*. We use a finite injury priority argument computable in $\emptyset'$. At stage s, we define a finite string $\sigma_s \in \{0,1\}^{<\omega}$ of length exactly s, letting $A_s = \{z < s : \sigma_s(z) = 1\}$ be the finite set of elements already enumerated into A.

Each requirement R_e can be in one of four states: *Initial*, *Working*, *Waiting*, or *Satisfied*. When in the *Waiting* state, R_e stores two parameters: a finite base set D_e and a length bound L_e. When in the *Working* state, R_e stores a target finite string $\tau_e \supset \sigma_s$.

Construction.

- **Stage 0:** $\sigma_0 = \emptyset$. All requirements are in the *Initial* state.
- **Stage $s+1$:** Let $x' = s$. Find the least $e \leq s$ such that R_e is not *Satisfied* and requires attention. R_e requires attention if one of the following conditions holds:

  C1 R_e is *Initial* and $\exists x \geq x' \exists y \notin A_s \cup \{x\}\, (y \in V_{e,x})$. Since $V_{e,x}$ is c.e. and A_s is a known finite set, this is a Σ_1^0 condition, which can be evaluated using $\emptyset'$.

  C2 R_e is *Initial* and **C1** is false.

  C3 R_e is *Waiting* (with parameters D_e, L_e). By the failure of **C1** when R_e entered the *Waiting* state, we know that $V_{e,x'} \subseteq D_e \cup \{x'\}$. Using $\emptyset'$, we can exactly compute the finite set $V_{e,x'}$. Condition **C3** holds if $x' \notin V_{e,x'}$ and the set $S = V_{e,x'} \cap D_e$ is "recorded". We say S is recorded if there exists some $x \in [L_e, x')$ such that $\sigma_s(x) = 1$, $x \notin V_{e,x}$, and $V_{e,x} \cap D_e = S$.

  C4 R_e is *Working* (with target string τ_e). This naturally holds as long as $x' < |\tau_e|$.

  If no such e exists, set $\sigma_{s+1} = \sigma_s \frown 1$ (to implicitly satisfy the positive requirements P_k), and proceed to the next stage. Otherwise, let e be the least such index, and act according to the condition satisfied:

  - **Action for C1:** Use $\emptyset'$ to find the lexicographically least pair (x, y) witnessing **C1**. Let τ_e be the unique extension of σ_s of length $\max(x, y)+1$ such that $\tau_e(x) = 1$, $\tau_e(y) = 0$ (if $y \geq x'$), and $\tau_e(z) = 0$ for all other $z \in [x', \max(x, y)+1) \setminus \{x, y\}$. Set $\sigma_{s+1} = \sigma_s \frown \tau_e(x')$. If $|\sigma_{s+1}| = |\tau_e|$, set R_e to *Satisfied*; otherwise, set R_e to *Working* with target τ_e. Initialize all R_i for $i > e$ (i.e., reset them to *Initial*).
  - **Action for C2:** Set R_e to *Waiting* with $D_e = A_s$ and $L_e = x'$. Set $\sigma_{s+1} = \sigma_s \frown 1$, and initialize all R_i for $i > e$.
  - **Action for C3:** Set $\sigma_{s+1} = \sigma_s \frown 0$ (so $x' \notin A$). Set R_e to *Satisfied*, and initialize all R_i for $i > e$.
  - **Action for C4:** Set $\sigma_{s+1} = \sigma_s \frown \tau_e(x')$. If $|\sigma_{s+1}| = |\tau_e|$, set R_e to *Satisfied*. Initialize all R_i for $i > e$.

We set $A = \bigcup_s A_s$. Since $|\sigma_s| = s$ exactly at each stage, the construction is well-defined. All decisions and searches at stage $s+1$ are computable from $\emptyset'$, hence $A \in \Delta_2^0$.

Verification. By standard finite injury, each R_e is initialized only finitely often. After its last initialization, R_e starts in *Initial*. It can act via **C1**, after which it acts finitely many times via **C4** (advancing bit-by-bit) until it becomes *Satisfied*. Alternatively, it acts via **C2** (entering *Waiting*). Once in the *Waiting* state, its parameters D_e, L_e are permanently fixed, and it may later act at most once via **C3**, becoming *Satisfied*. Thus, every requirement eventually stabilizes. It remains to verify that A is infinite. If infinitely many stages use the default action, then infinitely many 1's are appended, so A is infinite. Otherwise, infinitely many stages are devoted to actions of requirements. Since each fixed requirement acts only finitely many times, infinitely many distinct requirements act. For each such requirement, consider its first action after its last initialization. That action is either **C2**, which immediately writes a 1, or **C1**. In the latter case, the chosen target string τ_e contains some coordinate x with $\tau_e(x) = 1$, and since no further initialization of R_e occurs afterwards, that 1 is eventually written into the construction, either immediately at the **C1** stage or at a later **C4**

stage when τ_e is completed. Hence infinitely many 1's are written into the characteristic function of A, so A is infinite. Therefore every P_e is satisfied. To see that R_e is met, suppose for a contradiction that the family $(V_{e,x})_{x\in\omega}$ Q-reduces A to a coinfinite subset $B \subseteq A$. We analyze the final state of R_e:

- If R_e is *Satisfied* via **C1**/**C4**: The target string τ_e was fully written. We forced $x \in A$ and $y \notin A$, with $y \in V_{e,x}$. Since $B \subseteq A$, $y \notin B$, so $V_{e,x} \not\subseteq B$. However, $x \in A$, so the Q-reduction would require $V_{e,x} \subseteq B$. Contradiction.

- If R_e is *Satisfied* via **C3**: We forced $x' \notin A$, and there is some recorded element $x \in [L_e, x')$ with $x \in A$ such that $V_{e,x} \cap D_e = V_{e,x'} \cap D_e = S$. Since both $x, x' \geq L_e$ and **C1** globally failed for R_e when it entered *Waiting*, we have $V_{e,x} \subseteq D_e \cup \{x\}$ and $V_{e,x'} \subseteq D_e \cup \{x'\}$. Since $x \notin V_{e,x}$ and $x' \notin V_{e,x'}$, this implies exactly $V_{e,x} = S$ and $V_{e,x'} = S$. Since $x \in A$, the reduction implies $V_{e,x} \subseteq B$, so $S \subseteq B$. But $x' \notin A$, which implies $V_{e,x'} \not\subseteq B$, so $S \not\subseteq B$. Contradiction.

- If R_e remains *Waiting* forever: Then **C3** never triggers. By the failure of **C1** when R_e entered the *Waiting* state, we have

$$V_{e,x} \subseteq D_e \cup \{x\} \qquad \text{for every } x \geq L_e.$$

  For every $x > L_e$, the stage deciding whether x enters A evaluates **C3** for R_e; the only exceptional value is $x = L_e$, which was put into A when R_e entered the *Waiting* state via **C2**. This causes no problem. Indeed, if $x \geq L_e$, $x \in A$, and $x \notin V_{e,x}$, let

$$S_x = V_{e,x} \cap D_e.$$

  Since $x \notin V_{e,x}$ and $V_{e,x} \subseteq D_e \cup \{x\}$, we actually have $V_{e,x} = S_x \subseteq D_e$. For each fixed subset $S \subseteq D_e$, at most the least $x \geq L_e$ with

$$x \in A, \qquad x \notin V_{e,x}, \qquad V_{e,x} \cap D_e = S$$

  can belong to A: once such an x exists, the set S is recorded. If any later $x' > x$ existed with $x' \notin V_{e,x'}$ and $V_{e,x'} \cap D_e = S$, it would satisfy condition **C3**, causing R_e to act and become *Satisfied*. This contradicts the assumption that R_e remains *Waiting* for ever. Thus, no such x' can exist (and in particular, no such x' can belong to A). Hence there are at most $2^{|D_e|}$ many $x \geq L_e$ such that $x \in A$ and $x \notin V_{e,x}$. Since there are only finitely many $x < L_e$, it follows that only finitely many $x \in A$ satisfy $x \notin V_{e,x}$. Therefore, for almost all $x \in A$, we have $x \in V_{e,x}$. By the Q-reduction, $x \in A$ implies $V_{e,x} \subseteq B$, and hence $x \in B$ for almost all $x \in A$. This means that $A \setminus B$ is finite, contradicting that B is coinfinite in A.

In all cases, the reduction fails. Hence A is Q-introimmune. □

5 Enumeration reducibility: nonexistence at the Π^1_1 level and abstract existence

We now turn to enumeration reducibility [10]. In contrast with the previous reducibilities, the situation for $\leq_e$ is not arithmetical: we first show that no infinite Π^1_1 set is e-introimmune, and hence that there are no arithmetical or hyperarithmetical examples. We then show, however, that e-introimmune sets do exist in the unrestricted sense.

Recall that for sets $A, B \subseteq \omega$ one writes $A \leq_e B$ if there exists a computably enumerable set $W \subseteq \omega \times [\omega]^{<\omega}$ such that

$$x \in A \iff (\exists F \subseteq B \text{ finite})\ \langle x, F\rangle \in W.$$

For an infinite set $X = \{x_0 < x_1 < \cdots\}$, let $p_X(n) = x_n$ denote its principal function.

The next theorem gives a negative result for $\leq_e$ analogous in spirit to the nonexistence theorems of Jockusch and Simpson for Turing introimmunity [13, 15].

The key inputs are [11, Proposition 2.3], which characterizes Π^1_1 sets by uniform c.e. moduli, and [11, Proposition 2.1], which identifies uniform relative c.e. procedures with uniform enumeration operators.

Theorem 5.1. *Let $A \subseteq \omega$ be infinite and Π^1_1. Then there exists a set $B \subseteq A$ such that $|A \setminus B| = \infty$ and, moreover, there is a single enumeration operator Ψ satisfying*

$$\Psi(C) = A \qquad \textit{for every infinite } C \subseteq B.$$

In particular, $A \leq_e B$.

Proof. By [11, Proposition 2.3], there exist a function $f \in \omega^\omega$ and a relative c.e. operator Θ such that

$$\Theta(g) = A \qquad \text{for every total function } g \geq f.$$

Let p_A be the principal function of A. Since p_A is strictly increasing and unbounded, we may choose inductively integers

$$n_0 < n_1 < n_2 < \cdots$$

such that

$$p_A(n_s) \geq f(s) \qquad \text{and} \qquad n_{s+1} > n_s + 1$$

for every $s \in \omega$. Set

$$B = \{p_A(n_s) : s \in \omega\}.$$

Then $B \subseteq A$ is infinite and its principal function satisfies

$$p_B(s) = p_A(n_s) \geq f(s) \qquad \text{for all } s.$$

Also, $A \setminus B$ is infinite, because for each s we have $n_s + 1 < n_{s+1}$, hence

$$p_A(n_s + 1) \in A \setminus B.$$

We claim that A is uniformly c.e. in every infinite subset of B. Let $C \subseteq B$ be infinite. Then

$$p_C \geq p_B \geq f,$$

so

$$\Theta(p_C) = A.$$

To make the uniformity explicit, fix a c.e. set of axioms defining Θ, that is, a c.e. set

$$E \subseteq \omega^{<\omega} \times \omega$$

such that

$$x \in \Theta(g) \iff (\exists \sigma \prec g)\ (\sigma, x) \in E.$$

For each strictly increasing finite sequence

$$\sigma = (u_0 < \cdots < u_{k-1}) \in \omega^{<\omega}$$

(with the convention that $\tau_\emptyset = \emptyset$), let $\tau_\sigma \in 2^{<\omega}$ be the binary string of length $u_{k-1}+1$ whose 1's occur exactly at the positions $u_0, \ldots, u_{k-1}$. Define

$$\widehat{E} = \{(\tau_\sigma, x) : \sigma \text{ is strictly increasing and } (\sigma, x) \in E\},$$

and let $\widehat{\Theta}$ be the relative c.e. operator on set oracles determined by $\widehat{E}$. Then, for every infinite set $C \subseteq \omega$,

$$\widehat{\Theta}(C) = \Theta(p_C),$$

viewing C as its characteristic function, because for every strictly increasing σ, one has $\sigma \prec p_C$ if and only if $\tau_\sigma \prec C$. Therefore, for every infinite $C \subseteq B$,

$$\widehat{\Theta}(C) = \Theta(p_C) = A.$$

So A is uniformly c.e. in every infinite subset of B.

By [11, Proposition 2.1], there exists a single enumeration operator Ψ such that

$$\Psi(C) = A \qquad \text{for every infinite } C \subseteq B.$$

In particular, taking $C = B$, we obtain $A \leq_e B$. □

Corollary 5.2. *No infinite Π^1_1 set is e-introimmune. Consequently, there are no arithmetical e-introimmune sets and no hyperarithmetical e-introimmune sets.*

Proof. Let A be infinite and Π^1_1. By Theorem 5.1, there is a set $B \subseteq A$ such that $|A \setminus B| = \infty$ and $A \leq_e B$. Hence A is not e-introimmune. The final statement follows because every arithmetical or hyperarithmetical set is Δ^1_1, hence in particular Π^1_1. □

Remark 5.3. Theorem 5.1 actually yields more than the failure of e-introimmunity: after passing to a suitable coinfinite subset $B \subseteq A$, the set A is uniformly enumeration reducible from *every* infinite subset of B.

Thus e-introimmunity disappears already below the Π^1_1 level. It is therefore natural to ask whether the notion is vacuous or whether non-effective examples still exist. The next theorem shows that e-introimmune sets do exist, by an application of Soare's general existence theorem [16] to the countable family of enumeration operators.

Theorem 5.4. *There exists an e-introimmune set.*

Proof. Let $\{\Psi_e\}_{e\in\omega}$ be a standard effective enumeration of all enumeration operators, and view each Ψ_e as a total map from 2^ω to 2^ω.

We first observe that each Ψ_e is Borel measurable. Fix e and x, and let

$$U_{e,x} = \{X \in 2^\omega : x \in \Psi_e(X)\}.$$

If $E_e \subseteq \omega \times [\omega]^{<\omega}$ is the c.e. set of axioms defining Ψ_e, then

$$U_{e,x} = \bigcup_{\langle x, F\rangle \in E_e} \{X \in 2^\omega : F \subseteq X\},$$

hence $U_{e,x}$ is open. Therefore, if $[\sigma] \subseteq 2^\omega$ is a basic clopen cylinder, then

$$\Psi_e^{-1}([\sigma]) = \bigcap_{\substack{n<|\sigma| \\ \sigma(n)=1}} U_{e,n} \cap \bigcap_{\substack{n<|\sigma| \\ \sigma(n)=0}} (2^\omega \setminus U_{e,n}),$$

which is Borel. Thus Ψ_e is Borel measurable.

Applying Soare's theorem to the countable family $\{\Psi_e\}_{e\in\omega}$ and to $A = \omega$, we obtain an infinite set $B \subseteq \omega$ such that

$$(\forall e)(\forall S, T)\big[S, T \subseteq B \text{ infinite } \&\Psi_e(S) = T \to T \setminus S \text{ is finite}\big].$$

We claim that B is e-introimmune.

Suppose toward a contradiction that $B \leq_e C$ for some $C \subseteq B$ with $|B \setminus C| = \infty$. Let e be such that $\Psi_e(C) = B$.

If C is infinite, then taking $S = C$ and $T = B$ contradicts the displayed property, since $B \setminus C$ is infinite.

So C must be finite. But then B is computably enumerable, because

$$x \in B \iff (\exists F \subseteq C)\ \langle x, F\rangle \in E_e,$$

and there are only finitely many subsets of C. Choose any infinite coinfinite set $R \subseteq B$. Since B is c.e., the set of axioms $\{\langle x, \emptyset\rangle : x \in B\}$ is c.e., which defines an enumeration operator which ignores the oracle and simply enumerates B; hence $B \leq_e R$. Now R is infinite, so taking $S = R$ and $T = B$ again contradicts the displayed property, because $B \setminus R$ is infinite.

This contradiction shows that no such C exists. Therefore B is e-introimmune. □

Remark 5.5. Theorem 5.4 yields more than the existence of a single e-introimmune set. The set B obtained in the proof has the stronger property that for all infinite $S, T \subseteq B$, if $T \leq_e S$, then $T \setminus S$ is finite. In particular, the same argument shows that every infinite subset of B is itself e-introimmune. Combined with Corollary 5.2, this shows that enumeration reducibility admits e-introimmune sets only in a genuinely non-effective form: they do exist, but no infinite Π^1_1 set can be one.

6 Conclusions and Future Work

The aim of this paper has been to clarify how low introimmune sets can occur for several reducibilities arising in computability theory. For reducibilities strictly below Turing reducibility, we obtained three arithmetical existence results: a *wtt*-introimmune set in Π^0_1, a *bs*-introimmune set in Δ^0_2, and a D-introimmune set in Δ^0_2; consequently, a Δ^0_2 D^+-introimmune set exists as well. The first of these results answers the open question left in [5] and shows that Π^0_1 is the optimal arithmetical level for weak truth-table introimmunity, since Σ^0_1 sets cannot be immune.

We also analyzed the classical reducibility $\leq_Q$, showing that no infinite Π^0_1 set is Q-introimmune while a Δ^0_2 Q-introimmune set does exist. Thus the existence of Δ^0_2 Q-introimmune sets is best possible within the arithmetical hierarchy. Finally, for enumeration reducibility $\leq_e$, we obtained a qualitatively different picture: no infinite Π^1_1 set is e-introimmune, so there are no arithmetical, hyperarithmetical, or even Π^1_1 examples; nevertheless, by applying Soare's abstract theorem to the countable family of enumeration operators, we proved that e-introimmune sets do exist. Thus, in the enumeration case, introimmunity survives only in a genuinely non-effective form.

Taken together, these results considerably enlarge the current existence picture for introimmune sets. They show that the descriptive complexity of introimmune sets is highly sensitive to the

form of oracle access allowed by the reducibility. For $\leq_{wtt}$, the rigid computable use can still be handled inside a purely subtractive Π^0_1 construction by means of the dynamic spacing technique developed in Section 2. For $\leq_{bs}$ and $\leq_D$, by contrast, no computable a priori use bound is available, and the required spacing must be computed with the aid of the halting problem; this is why our present constructions for these reducibilities live in Δ^0_2 rather than in Π^0_1. For $\leq_Q$, a bit-by-bit finite-extension construction yields a sharp Δ^0_2 existence theorem together with a Π^0_1 obstruction. For $\leq_e$, the phenomenon is no longer arithmetical at all: the nonexistence theorem is obtained from the descriptive-set-theoretic characterization of Π^1_1 sets by uniform c.e. moduli, whereas abstract existence follows from Soare's theorem for countable families of Borel maps.

A natural next problem is to understand the degree-theoretic distribution of Π^0_1 *wtt*-introimmune sets. It is known [6] that every non-zero c.e. degree computes a Π^0_1 *m*-introimmune set. It is therefore natural to ask whether an analogous ubiquity phenomenon holds for *wtt*-introimmunity. Our present construction suggests serious obstacles to a straightforward permitting argument: the dynamic spacing method relies on preserving carefully prepared gaps, whereas external permissions may arrive at stages that allow an opponent to destroy backup configurations infinitely often. This leads us to conjecture that, unlike the many-one case, there are non-zero c.e. Turing degrees that compute no Π^0_1 *wtt*-introimmune set.

Another natural direction is to sharpen the results obtained for the other reducibilities studied here. For $\leq_{bs}$ and $\leq_D$, the main open question is whether the present Δ^0_2 existence results can be lowered to Π^0_1. For $\leq_Q$, since the minimal arithmetical existence level has now been identified, it would be interesting to investigate what additional structural restrictions can be imposed on Δ^0_2 Q-introimmune sets. For $\leq_e$, where no infinite Π^1_1 example can exist, a natural problem is to locate the least possible descriptive complexity of *e*-introimmune sets above the Π^1_1 level.

These problems suggest that introimmunity remains closely connected with the fine structure of reducibilities, and that even small changes in the definition of reducibility may lead to markedly different arithmetical and descriptive existence phenomena.

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